UC Berkeley

Department of Electrical Engineering and Computer Sciences

EECS 127 - EECS 227AT. Optimization Models in Engineering

Lecture Note 10

Fall 2025

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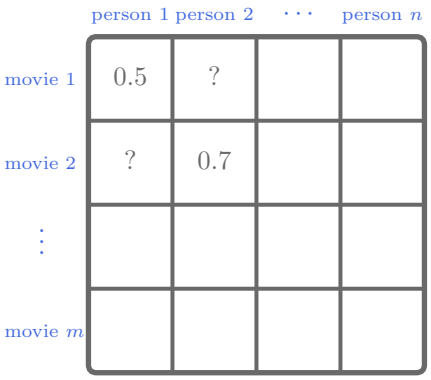
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Assume $\mathbf{A} \in \mathbb{R}^{m \times n}$ is approximated by a matrix $\mathbf{B}$ such that $\text{rank}(\mathbf{B}) \leq k$:

$$\mathbf{B} = \boldsymbol{c}^{(1)} \left(\boldsymbol{d}^{(1)}\right)^{\mathrm{T}} + \cdots + \boldsymbol{c}^{(k)} \left(\boldsymbol{d}^{(k)}\right)^{\mathrm{T}}.$$

- The amount of data in $\mathbf{B}$ is $k(m+n)$, while the amount of data in $\mathbf{A}$ is mn.
- If k is close to 1, this leads to a huge reduction in data.
- In Netflix, it happens that k is expected to be close to 1, meaning that there are a few (k) columns of $\mathbf{B}$ such that every other column is a linear combination of these selected columns.
- This means there are a few (k) people such that the movie preferences of every other person is a linear combination of preferences of these selected people.
- For Netflix, $\mathbf{A}$ is not known since not everyone has watched all movies:



- When Netflix assumes that $\mathbf{A}$ is low-rank, it can find unknown entries via a machine learning problem, named *matrix completion* (to be discussed later).
- After learning $\mathbf{A}$, Netflix learns preferences of people for movies they haven't watched, so it suggests them movies they will like if they watch them.

Given $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $k \leq \min(m, n)$, define:

$$\mathbf{A}_k = \sigma_1 \boldsymbol{u}^{(1)} \left(\boldsymbol{v}^{(1)}\right)^{\mathrm{T}} + \sigma_2 \boldsymbol{u}^{(2)} \left(\boldsymbol{v}^{(2)}\right)^{\mathrm{T}} + \cdots + \sigma_k \boldsymbol{u}^{(k)} \left(\boldsymbol{v}^{(k)}\right)^{\mathrm{T}},$$

where $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_k$ are the singular values of $\mathbf{A}$ and $\boldsymbol{u}^{(i)}, \boldsymbol{v}^{(i)}$ are the corresponding singular vectors.

- If $k = \min(m, n)$, then $\mathbf{A}_k = \mathbf{A}$.
- $\mathbf{A}_k$ has rank at most k and it is based on the top k singular values of $\mathbf{A}$ and their corresponding left and right singular vectors.

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Theorem 1 (Eckart-Young-Mirsky theorem)**.** $\mathbf{B} = \mathbf{A}_k$ *is an optimal solution to both optimization problems:*

$$\min_{\mathbf{B}\in\mathbb{R}^{m\times n}} \|\mathbf{A}-\mathbf{B}\|_{\mathrm{F}} \quad s.t.\ \operatorname{rank}(\mathbf{B}) \le k; \tag{1}$$

$$\min_{\mathbf{B}\in\mathbb{R}^{m\times n}} \|\mathbf{A}-\mathbf{B}\|_{2} \quad s.t.\ \operatorname{rank}(\mathbf{B}) \le k. \tag{2}$$

Also, the optimal solution is unique iff $\sigma_k \neq \sigma_{k+1}$ *(recall* $\sigma_1 \ge \sigma_2 \ge \cdots \ge \sigma_k \ge \sigma_{k+1} \ge \cdots$*).*

(Note: there is a proof for part (1) in the textbook. It gives good intuition but is not 100% mathematically correct.)

Proof of part (1)*.* Let r denote the rank of $\mathbf{A}$. Then

$$\begin{aligned}\mathbf{A}-\mathbf{A}_k &= \sum_{i=1}^{r}\sigma_i \boldsymbol{u}^{(i)}\left(\boldsymbol{v}^{(i)}\right)^{\mathrm{T}} - \sum_{i=1}^{k}\sigma_i \boldsymbol{u}^{(i)}\left(\boldsymbol{v}^{(i)}\right)^{\mathrm{T}} \\ &= \sum_{i=k+1}^{r}\sigma_i \boldsymbol{u}^{(i)}\left(\boldsymbol{v}^{(i)}\right)^{\mathrm{T}} + \sum_{i=1}^{k} 0\times \boldsymbol{u}^{(i)}\left(\boldsymbol{v}^{(i)}\right)^{\mathrm{T}}.\end{aligned}$$

Here $\sigma_1,\ldots,\sigma_r,0,\ldots,0$ are the singular values of $\mathbf{A}$, and the above formula says $\sigma_{k+1},\ldots,\sigma_r$, $0,\ldots,0$ are the singular values of $\mathbf{A}-\mathbf{A}_k$. Its norm is given by

$$\begin{aligned}\|\mathbf{A}-\mathbf{A}_k\|_{\mathrm{F}}^2 &= \operatorname{trace}\left((\mathbf{A}-\mathbf{A}_k)^{\mathrm{T}}(\mathbf{A}-\mathbf{A}_k)\right) \\ &= \operatorname{trace}\left(\left(\sum_{i=k+1}^{r}\sigma_i \boldsymbol{v}^{(i)}\left(\boldsymbol{u}^{(i)}\right)^{\mathrm{T}}\right)\left(\sum_{j=k+1}^{r}\sigma_j \boldsymbol{u}^{(j)}\left(\boldsymbol{v}^{(j)}\right)^{\mathrm{T}}\right)\right) \\ &= \operatorname{trace}\left(\sum_{i=k+1}^{r}\sum_{j=k+1}^{r}\sigma_i\sigma_j \boldsymbol{v}^{(i)}\left(\boldsymbol{u}^{(i)}\right)^{\mathrm{T}}\boldsymbol{u}^{(j)}\left(\boldsymbol{v}^{(j)}\right)^{\mathrm{T}}\right).\end{aligned}$$

Notice that

$$\left(\boldsymbol{u}^{(i)}\right)^{\mathrm{T}}\boldsymbol{u}^{(j)} = \begin{cases}1, & \text{if } i=j,\\ 0, & \text{if } i\neq j\end{cases} \quad \text{and} \quad \operatorname{trace}\left(\boldsymbol{v}^{(i)}\left(\boldsymbol{v}^{(i)}\right)^{\mathrm{T}}\right) = \|\boldsymbol{v}^{(i)}\|^2 = 1,$$

we have

$$\begin{aligned}\|\mathbf{A}-\mathbf{A}_k\|_{\mathrm{F}}^2 &= \operatorname{trace}\left(\sum_{i=k+1}^{r}\sigma_i^2 \boldsymbol{v}^{(i)}\left(\boldsymbol{v}^{(i)}\right)^{\mathrm{T}}\right) \\ &= \sum_{i=k+1}^{r}\sigma_i^2\cdot\operatorname{trace}\left(\boldsymbol{v}^{(i)}\left(\boldsymbol{v}^{(i)}\right)^{\mathrm{T}}\right) \\ &= \sum_{i=k+1}^{r}\sigma_i^2.\end{aligned}$$

(If $k=r$, by convention, we define $\sigma_{k+1}=\sigma_{r+1}$ to be zero.)

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To show that $\mathbf{A}_k$ is a solution to (1), we need to prove that

$$\|\mathbf{A}-\mathbf{B}\|_{\mathrm{F}}^2 \geq \sum_{i=k+1}^{r} \sigma_i^2, \quad \forall \mathbf{B} \text{ s.t. } \operatorname{rank}(\mathbf{B}) \leq k. \tag{3}$$

We need two lemmas:

Lemma 1. *Given a square matrix* $\mathbf{X}$*, we have* $\operatorname{trace}(\mathbf{X})$ *is equal to the sum of all eigenvalues of* $\mathbf{X}$*.*

Lemma 2. *Given two matrices* $\mathbf{Y}, \mathbf{Z} \in \mathbb{R}^{m\times n}$*, for every* $i, j \geq 1$*, we have*

$$\sigma_i(\mathbf{Y}) + \sigma_j(\mathbf{Z}) \geq \sigma_{i+j-1}(\mathbf{Y}+\mathbf{Z}).$$

(By convention, $\sigma_{i+j-1} = 0$ *if* $i+j-1 > \min(m,n)$*.)*

From Lemma 1, we have:

$$\begin{aligned}\|\mathbf{X}\|_{\mathrm{F}}^2 = \operatorname{trace}(\mathbf{X}\mathbf{X}^{\mathrm{T}}) &= \lambda_1(\mathbf{X}\mathbf{X}^{\mathrm{T}}) + \cdots + \lambda_m(\mathbf{X}\mathbf{X}^{\mathrm{T}}) \\ &= \sigma_1^2(\mathbf{X}) + \cdots + \sigma_r^2(\mathbf{X}).\end{aligned}$$

(Recall: the singular values of $\mathbf{X}$ are the square roots of the eigenvalues of $\mathbf{X}\mathbf{X}^{\mathrm{T}}$.)
If $\operatorname{rank}(\mathbf{B}) \leq k$, we have

$$\begin{aligned}\|\mathbf{A}-\mathbf{B}\|_{\mathrm{F}}^2 &= \sum_{i=1}^{\min(m,n)} \sigma_i^2(\mathbf{A}-\mathbf{B}) \qquad \text{(due to Lemma 1)} \\ &= \sum_{i=1}^{\min(m,n)} \big(\sigma_i(\mathbf{A}-\mathbf{B}) + \sigma_{k+1}(\mathbf{B})\big)^2. \end{aligned} \tag{4}$$

(This is true since $\sigma_{k+1}(\mathbf{B}) = 0$ since $\mathbf{B}$ has rank at most k and so has at most k non-zero singular values.)
By Lemma 2 and Equation (4):

$$\begin{aligned}\|\mathbf{A}-\mathbf{B}\|_{\mathrm{F}}^2 &\geq \sum_{i=1}^{\min(m,n)} \sigma_{i+(k+1)-1}^2\big((\mathbf{A}-\mathbf{B})+\mathbf{B}\big) \\ &= \sum_{i=1}^{\min(m,n)} \sigma_{i+k}^2(\mathbf{A}) \\ &= \sum_{j=k+1}^{r} \sigma_j^2(\mathbf{A}),\end{aligned}$$

where the last step is because $\sigma_j = 0$ if $j > r$. Therefore, (3) is proven.
The uniqueness is not proven in class. ■

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Before proving part (2), let's get some intuition behind part (1).
When $\mathbf{A}_k$ is the best approximation of $\mathbf{A}$, we should measure how much we lose through approximation:

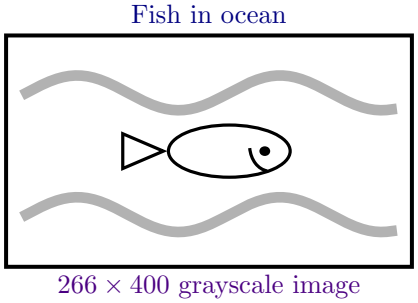
$$e_k = \frac{\overbrace{\boxed{\|\mathbf{A}-\mathbf{A}_k\|_{\mathrm{F}}^2}}^{\text{residue}}}{\underbrace{\boxed{\|\mathbf{A}\|_{\mathrm{F}}^2}}_{\text{true matrix}}} = \frac{\sigma_{k+1}^2+\cdots+\sigma_r^2}{\sigma_1^2+\cdots+\sigma_r^2} =: 1-\eta_k,$$

where

$$\eta_k = \frac{\|\mathbf{A}_k\|_{\mathrm{F}}^2}{\|\mathbf{A}\|_{\mathrm{F}}^2} = \frac{\sigma_1^2+\cdots+\sigma_k^2}{\sigma_1^2+\cdots+\sigma_r^2}.$$

η_k indicates how much total variance (in Frobenius norm) in $\mathbf{A}$ is captured by $\mathbf{A}_k$. Smaller e_k indicates better approximation.

Example 1 (Example 5.2 in textbook)**.** Assume we have a grayscale image $\mathbf{A} \in \mathbb{R}^{266\times 400}$ with $\operatorname{rank}(\mathbf{A}) = 266$:



For $k = 9$, we already have $\eta_k = 0.96$. Thus, using 9 significant singular values out of 266 already gives a good approximation:

$$\underbrace{\mathbf{A}}_{266\times 400} \approx \begin{bmatrix} \cdots \end{bmatrix}_{266\times 9} \begin{bmatrix} \quad \cdots \quad \end{bmatrix}_{9\times 400}.$$

■

Proof of part (2). We have

$$\|\mathbf{A}-\mathbf{A}_k\|_2 = \left\| \sum_{i=k+1}^{r} \sigma_i \boldsymbol{u}^{(i)} \left(\boldsymbol{v}^{(i)}\right)^{\mathrm{T}} \right\|_2,$$

where σ_i is the singular value of $\mathbf{A}$.
Recall:

(i) Singular values of $\sum_{i=k+1}^{r} \sigma_i \boldsymbol{u}^{(i)} (\boldsymbol{v}^{(i)})^{\mathrm{T}}$ are $\sigma_{k+1} \geq \sigma_{k+2} \geq \cdots \geq \sigma_r \geq 0 \geq \cdots$.

(ii) $\|\text{matrix}\|_2$ is equal to the maximum singular value of the matrix.

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Thus,

$$\|\mathbf{A} - \mathbf{A}_k\|_2 = \sigma_{k+1}.$$

It remains to show that

$$\|\mathbf{A} - \mathbf{B}\|_2 \geq \sigma_{k+1}(\mathbf{A}), \quad \forall \mathbf{B} \text{ s.t. } \operatorname{rank}(\mathbf{B}) \leq k.$$

Since $\operatorname{rank}(\mathbf{B}) \leq k$, there exist matrices $\mathbf{C} \in \mathbb{R}^{m\times k}$ and $\mathbf{D} \in \mathbb{R}^{n\times k}$ such that $\mathbf{B} = \mathbf{C}\mathbf{D}^{\mathrm{T}}$. Consider the SVD of $\mathbf{A}$: $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^{\mathrm{T}}$. Notice that:

- $\operatorname{span}(\boldsymbol{d}^{(1)}, \ldots, \boldsymbol{d}^{(k)})$ has dimension at most k;
- $\operatorname{span}(\boldsymbol{v}^{(1)}, \ldots, \boldsymbol{v}^{(k)}, \boldsymbol{v}^{(k+1)})$ has dimension $k+1$ (because $\boldsymbol{v}^{(i)}$'s are linearly independent).

Therefore, there exists a vector $\boldsymbol{w} \in \operatorname{span}(\boldsymbol{v}^{(1)}, \ldots, \boldsymbol{v}^{(k+1)})$ that is orthogonal to $\operatorname{span}(\boldsymbol{d}^{(1)}, \ldots, \boldsymbol{d}^{(k)})$. You can prove this claim by applying the rank-nullity theorem on the following map:

$$\begin{aligned} f\colon \operatorname{span}\left(\boldsymbol{v}^{(1)}, \ldots, \boldsymbol{v}^{(k+1)}\right) &\to \mathbb{R}^k \\ \boldsymbol{w} &\mapsto \left[\left(\boldsymbol{d}^{(1)}\right)^{\mathrm{T}} \boldsymbol{w}, \ldots, \left(\boldsymbol{d}^{(k)}\right)^{\mathrm{T}} \boldsymbol{w}\right]^{\mathrm{T}}. \end{aligned}$$

Let

$$\boldsymbol{w} = \alpha_1 \boldsymbol{v}^{(1)} + \cdots + \alpha_{k+1} \boldsymbol{v}^{(k+1)}$$

for some $\alpha_1, \ldots, \alpha_{k+1}$, such that

$$\left(\boldsymbol{d}^{(i)}\right)^{\mathrm{T}} \boldsymbol{w} = 0, \ i = 1, \ldots, k \quad \Rightarrow \quad \mathbf{D}^{\mathrm{T}} \boldsymbol{w} = \mathbf{0}.$$

We have

$$\|\mathbf{A} - \mathbf{B}\|_2 = \max_{\boldsymbol{x}\in\mathbb{R}^n} \frac{\|(\mathbf{A} - \mathbf{B})\boldsymbol{x}\|}{\|\boldsymbol{x}\|} \geq \frac{\|(\mathbf{A} - \mathbf{B})\boldsymbol{w}\|}{\|\boldsymbol{w}\|}.$$

(The inequality is because $\boldsymbol{w}$ is arbitrary so it cannot be better than the optimal $\boldsymbol{x}$.) Now,

$$\begin{aligned} \|(\mathbf{A} - \mathbf{B})\boldsymbol{w}\|^2 &= \left((\mathbf{A} - \mathbf{B})\boldsymbol{w}\right)^{\mathrm{T}} \left((\mathbf{A} - \mathbf{B})\boldsymbol{w}\right) \\ &= \boldsymbol{w}^{\mathrm{T}} \left(\mathbf{A}^{\mathrm{T}} - \mathbf{D}\mathbf{C}^{\mathrm{T}}\right) \left(\mathbf{A} - \mathbf{C}\mathbf{D}^{\mathrm{T}}\right) \boldsymbol{w} \\ &= \left(\boldsymbol{w}^{\mathrm{T}}\mathbf{A}^{\mathrm{T}} - \underbrace{\boldsymbol{w}^{\mathrm{T}}\mathbf{D}}_{=\mathbf{0}} \mathbf{C}^{\mathrm{T}}\right) \left(\mathbf{A}\boldsymbol{w} - \mathbf{C} \underbrace{\mathbf{D}^{\mathrm{T}}\boldsymbol{w}}_{=\mathbf{0}}\right) \\ &= \boldsymbol{w}^{\mathrm{T}}\mathbf{A}^{\mathrm{T}}\mathbf{A}\boldsymbol{w}. \end{aligned} \tag{5}$$

Also, since $(\boldsymbol{v}^{(i)})^{\mathrm{T}}\boldsymbol{v}^{(j)} = \begin{cases} 0, & \text{if } i \neq j, \\ 1, & \text{if } i = j, \end{cases}$

$$\begin{aligned} \mathbf{A}\boldsymbol{w} &= \left(\sum_{i=1}^{r} \sigma_i \boldsymbol{u}^{(i)} \left(\boldsymbol{v}^{(i)}\right)^{\mathrm{T}}\right) \left(\sum_{j=1}^{k+1} \alpha_j \boldsymbol{v}^{(j)}\right) \\ &= \sum_{j=1}^{k+1} \sigma_j \alpha_j \boldsymbol{u}^{(j)}. \end{aligned}$$

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Substituting this into Equation (5), we get

$$\begin{aligned}\boldsymbol{w}^{\mathrm{T}}\mathbf{A}^{\mathrm{T}}\mathbf{A}\boldsymbol{w} &= \left(\sum_{j=1}^{k+1}\sigma_j\alpha_j\boldsymbol{u}^{(j)}\right)^{\mathrm{T}}\left(\sum_{j=1}^{k+1}\sigma_j\alpha_j\boldsymbol{u}^{(j)}\right)\\ &= \sum_{j=1}^{k+1}\sigma_j^2\alpha_j^2\\ &\geq \sigma_{k+1}^2\underbrace{\boxed{\sum_{j=1}^{k+1}\alpha_j^2}}_{=\|\boldsymbol{w}\|^2},\end{aligned}$$

where the second equality is because

$$\left(\boldsymbol{u}^{(i)}\right)^{\mathrm{T}}\boldsymbol{u}^{(j)} = \begin{cases}0, & \text{if } i \neq j,\\ 1, & \text{if } i = j,\end{cases}$$

and the last inequality is because $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_{k+1}$. Therefore,

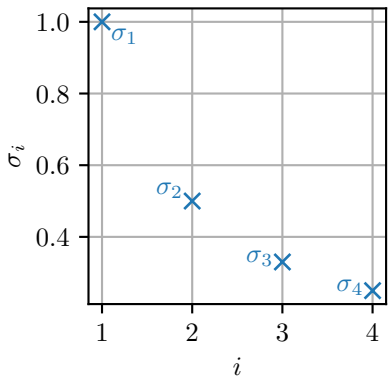
$$\|\mathbf{A}-\mathbf{B}\|_2 \geq \frac{\|(\mathbf{A}-\mathbf{B})\boldsymbol{w}\|}{\|\boldsymbol{w}\|} \geq \frac{\sqrt{\sigma_{k+1}^2\|\boldsymbol{w}\|^2}}{\|\boldsymbol{w}\|} = \sigma_{k+1}.$$

■

Error of approximating $\mathbf{A}$ by $\mathbf{A}_k$:

$$\frac{\|\mathbf{A}-\mathbf{A}_k\|_2}{\|\mathbf{A}\|_2} = \frac{\sigma_{k+1}}{\sigma_1}.$$

- To find the right low-rank approximation, we draw σ_i as a function of i:

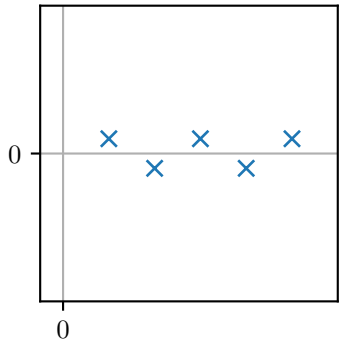


- Find the first k such that σ_{k+1} is sufficiently small compared to σ_1.
- Cut the tail of singular values to approximate $\mathbf{A}$.

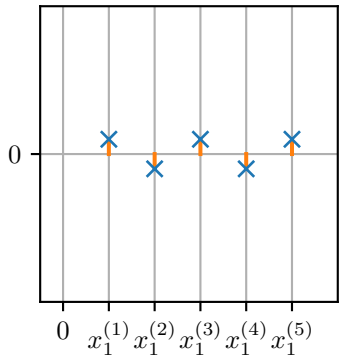
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Principal component analysis (PCA).

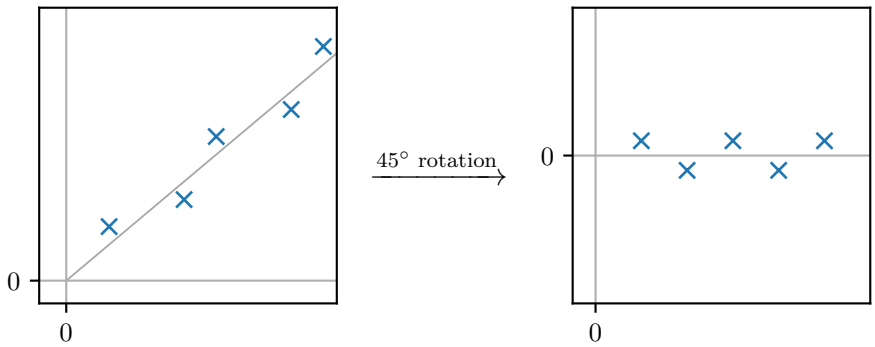
Example 2 (Motivating example)**.** Consider m points $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(m)}$ in $\mathbb{R}^2$ such that they are all close to the horizontal axis:



This means there is not much variation along the vertical axis and we can focus on the projection of these points on the horizontal axis:



Now, assume that $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(m)} \in \mathbb{R}^2$ are all close to the 45° line:

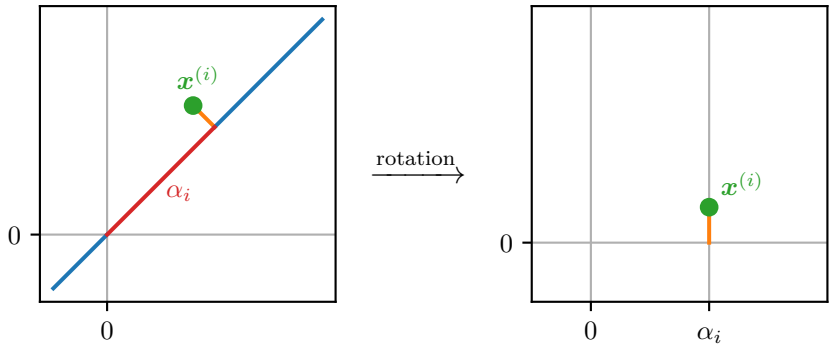


A 45° rotation converts this to the previous problem.
Alternatively, we consider the 45° line. Consider a point (direction) $\boldsymbol{z}$ on the line of length 1, then

$$\text{line} = \{\alpha \boldsymbol{z} \mid \alpha \in \mathbb{R}\}.$$

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Let the projection of $\boldsymbol{x}^{(i)}$ on the line be $\alpha_i \boldsymbol{z}$. Then, $\boldsymbol{x}^{(i)} \in \mathbb{R}^2$ is mapped into a scalar $\alpha_i \in \mathbb{R}$.



α_i acts like the coordinate along the 45° line. ■

To generalize this, consider data points $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(m)} \in \mathbb{R}^n$.

- Define the average $\tilde{\boldsymbol{x}} = \frac{1}{m}(\boldsymbol{x}^{(1)} + \cdots + \boldsymbol{x}^{(m)})$.
- Define the centered data points $\tilde{\boldsymbol{x}}^{(i)} = \boldsymbol{x}^{(i)} - \tilde{\boldsymbol{x}}$, $i = 1, \ldots, m$.
- **Question:** find a direction $\boldsymbol{z}$ (of length 1) along which $\tilde{\boldsymbol{x}}^{(1)}, \ldots, \tilde{\boldsymbol{x}}^{(m)}$ have maximum variation (in the previous example that corresponded to the 45° line).
- Define line $L = \{\alpha \boldsymbol{z} \mid \alpha \in \mathbb{R}\}$.
- The projection of $\tilde{\boldsymbol{x}}^{(i)}$ on L is $\alpha_i \boldsymbol{z}$, where

$$\alpha_i = \frac{\langle \tilde{\boldsymbol{x}}^{(i)}, \boldsymbol{z} \rangle}{\underbrace{\|\boldsymbol{z}\|}_{=1}} = \left(\tilde{\boldsymbol{x}}^{(i)}\right)^{\mathrm{T}} \boldsymbol{z} = \boldsymbol{z}^{\mathrm{T}} \tilde{\boldsymbol{x}}^{(i)}.$$

- Define the total square variation: $\alpha_1^2 + \cdots + \alpha_m^2$.
  (Since α_i could be positive or negative, we raise it to power 2.)

$$\sum_{i=1}^{m} \alpha_i^2 = \sum_{i=1}^{m} \left(\left(\tilde{\boldsymbol{x}}^{(i)}\right)^{\mathrm{T}} \boldsymbol{z}\right)^2 = \sum_{i=1}^{m} \left(\boldsymbol{z}^{\mathrm{T}} \tilde{\boldsymbol{x}}^{(i)}\right) \left(\left(\tilde{\boldsymbol{x}}^{(i)}\right)^{\mathrm{T}} \boldsymbol{z}\right) = \boldsymbol{z}^{\mathrm{T}} \tilde{\mathbf{X}} \tilde{\mathbf{X}}^{\mathrm{T}} \boldsymbol{z},$$

  where $\tilde{\mathbf{X}} = \begin{bmatrix} \tilde{\boldsymbol{x}}^{(1)} & \cdots & \tilde{\boldsymbol{x}}^{(m)} \end{bmatrix}$ is the centered data matrix.

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